Advanced Spam Detection System using Transformers

This project implements an advanced spam detection system powered by state-of-the-art transformer architectures such as RoBERTa and DeBERTa. The model leverages contextual embeddings and self-attention mechanisms to accurately classify text messages as spam or non-spam.

```
!pip install transformers datasets scikit-learn torch
```

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```

```
import transformers
print(transformers.__version__)
```

```
5.2.0
```

```
!pip install --upgrade transformers
```

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Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.2.0)
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```

```python
import torch
import pandas as pd
import numpy as np

from torch.utils.data import Dataset, DataLoader
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import  get_linear_schedule_with_warmup
from torch.optim import AdamW

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
```

```python
df = pd.read_csv("spam.csv")   # Must contain: text, label
df = df[['text', 'label']]

train_texts, val_texts, train_labels, val_labels = train_test_split(
    df['text'].values,
    df['label'].values,
    test_size=0.2,
    random_state=42,
    stratify=df['label']
)
```

```python
model_name = "roberta-base"
tokenizer = AutoTokenizer.from_pretrained(model_name)
```

```python
model_name = "roberta-base"
tokenizer = AutoTokenizer.from_pretrained(model_name)

class SpamDataset(Dataset):
    def __init__(self, texts, labels):
        self.encodings = tokenizer(
            texts.tolist(),
            truncation=True,
            padding=True,
            max_length=128
        )
        self.labels = labels

    def __getitem__(self, idx):
         item = {key: torch.tensor(val[idx]) for key, val in self.encodings.items()}

         label = self.labels[idx]
         if isinstance(label, str):
             label = 1 if label == "spam" else 0

         item["labels"] = torch.tensor(label)
         return item


    def __len__(self):
        return len(self.labels)
```

```
train_dataset = SpamDataset(train_texts, train_labels)
val_dataset = SpamDataset(val_texts, val_labels)

train_loader = DataLoader(train_dataset, batch_size=16, shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=16)
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

model = AutoModelForSequenceClassification.from_pretrained(model_name, num_labels=2)
model.to(device)
```

Loading weights: 100% 197/197 [00:00<00:00, 526.62it/s, Materializing param=roberta.encoder.layer.11.output.dense.weight]

```
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                              | Status     |
---------------------------------+------------+-
lm_head.layer_norm.weight        | UNEXPECTED |
roberta.embeddings.position_ids  | UNEXPECTED |
lm_head.bias                     | UNEXPECTED |
lm_head.dense.bias               | UNEXPECTED |
lm_head.layer_norm.bias          | UNEXPECTED |
lm_head.dense.weight             | UNEXPECTED |
classifier.out_proj.weight       | MISSING    |
classifier.dense.weight          | MISSING    |
classifier.out_proj.bias         | MISSING    |
classifier.dense.bias            | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream
RobertaForSequenceClassification(
  (roberta): RobertaModel(
    (embeddings): RobertaEmbeddings(
      (word_embeddings): Embedding(50265, 768, padding_idx=1)
      (token_type_embeddings): Embedding(1, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
      (position_embeddings): Embedding(514, 768, padding_idx=1)
    )
    (encoder): RobertaEncoder(
      (layer): ModuleList(
        (0-11): 12 x RobertaLayer(
          (attention): RobertaAttention(
            (self): RobertaSelfAttention(
              (query): Linear(in_features=768, out_features=768, bias=True)
              (key): Linear(in_features=768, out_features=768, bias=True)
              (value): Linear(in_features=768, out_features=768, bias=True)
              (dropout): Dropout(p=0.1, inplace=False)
            )
            (output): RobertaSelfOutput(
              (dense): Linear(in_features=768, out_features=768, bias=True)
              (LayerNorm): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
              (dropout): Dropout(p=0.1, inplace=False)
            )
          )
          (intermediate): RobertaIntermediate(
            (dense): Linear(in_features=768, out_features=3072, bias=True)
            (intermediate_act_fn): GELUActivation()
          )
          (output): RobertaOutput(
            (dense): Linear(in_features=3072, out_features=768, bias=True)
            (LayerNorm): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
        )
      )
    )
  )
  (classifier): RobertaClassificationHead(
    (dense): Linear(in_features=768, out_features=768, bias=True)
    (dropout): Dropout(p=0.1, inplace=False)
    (out_proj): Linear(in_features=768, out_features=2, bias=True)
  )
)
```

```
optimizer = AdamW(model.parameters(), lr=2e-5)

num_epochs = 3
total_steps = len(train_loader) * num_epochs

scheduler = get_linear_schedule_with_warmup(
    optimizer,
    num_warmup_steps=0,
    num_training_steps=total_steps
)
```

```
best_val_loss = float('inf')

for epoch in range(num_epochs):
    model.train()
    total_loss = 0

    for batch in train_loader:
        batch = {k: v.to(device) for k, v in batch.items()}

        outputs = model(**batch)
        loss = outputs.loss
        total_loss += loss.item()

        loss.backward()
        optimizer.step()
        scheduler.step()
        optimizer.zero_grad()

    avg_train_loss = total_loss / len(train_loader)

    # Validation
    model.eval()
    val_loss = 0
    predictions, true_labels = [], []

    with torch.no_grad():
        for batch in val_loader:
            batch = {k: v.to(device) for k, v in batch.items()}
            outputs = model(**batch)

            val_loss += outputs.loss.item()
            logits = outputs.logits

            preds = torch.argmax(logits, dim=1).cpu().numpy()
            labels = batch["labels"].cpu().numpy()

            predictions.extend(preds)
            true_labels.extend(labels)

    avg_val_loss = val_loss / len(val_loader)
    acc = accuracy_score(true_labels, predictions)

    print(f"\nEpoch {epoch+1}")
    print(f"Train Loss: {avg_train_loss:.4f}")
    print(f"Val Loss: {avg_val_loss:.4f}")
    print(f"Val Accuracy: {acc:.4f}")

    # Early stopping
    if avg_val_loss < best_val_loss:
        best_val_loss = avg_val_loss
        torch.save(model.state_dict(), "best_model.pt")
```

```
Epoch 1
Train Loss: 0.6895
Val Loss: 0.6491
Val Accuracy: 0.8333

Epoch 2
Train Loss: 0.6439
Val Loss: 0.6246
Val Accuracy: 0.8333

Epoch 3
Train Loss: 0.6307
Val Loss: 0.6146
Val Accuracy: 0.8333
```

```
def predict(text):
    model.eval()

    inputs = tokenizer(
        text,
        return_tensors="pt",
        truncation=True,
        padding=True,
        max_length=128
    ).to(device)

    with torch.no_grad():
        outputs = model(**inputs)
        logits = outputs.logits
        probs = torch.softmax(logits, dim=1)
        pred = torch.argmax(probs, dim=1).item()

    return {
        "prediction": "SPAM" if pred == 1 else "NOT SPAM",
        "confidence": float(probs[0][pred])
    }
```

```
result = predict("Congratulations! You won a free iPhone. Click here now!")

print(result)
```

```
texts = [
    "Hello how are you?",
    "Claim your reward now!!!"
]

for t in texts:
    print(predict(t))
```

```
{'prediction': 'NOT SPAM', 'confidence': 0.5587450265884399}
{'prediction': 'NOT SPAM', 'confidence': 0.5598067045211792}
```